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UF304

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UF304 / UF308

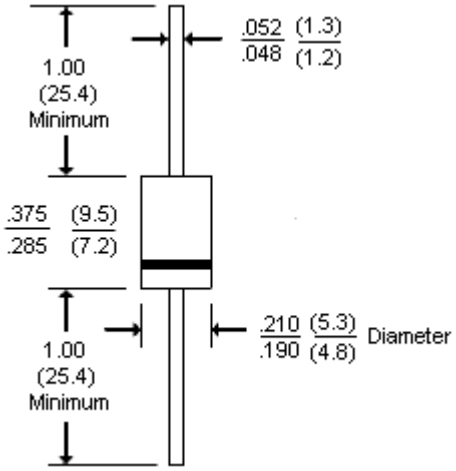
3A Ultra Fast Switching Rectifiers

multicomp

High Efficiency



DO-201AD



Dimensions : Inches (Millimetres)

Features:

- Fast reverse recovery time, t_{rr}.
- Low forward voltage, V_F.
- Low cost axial packages.
- Void-free plastic in DO-201AD package.
- 3.0 Ampere operation at T_A = 55°C with no thermal runaway.
- Exceeds environmental standards of MIL-S-19500/228.
- Ultra fast switching for high efficiency.

Mechanical Data:

Case	: Moulded plastic.
Terminals	: Axial leads, solderable per MIL-STD-202, Method 208.
Polarity	: Band denotes cathode.
Mounting Position	: Any.

Maximum Ratings and Electrical Characteristics:

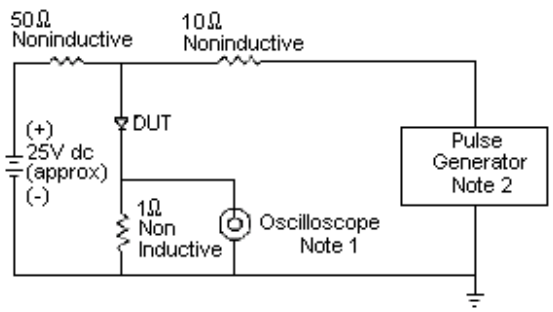
Ratings at 25°C ambient temperature unless otherwise specified.
Resistive or inductive load, 60Hz.

Parameters	UF304	UF308	Units
Peak reverse voltage, repetitive; V_{RM}	400	800	V
Maximum RMS voltage	280	560	
DC blocking voltage; V_R	400	800	
Average forward current, I_o at T_A = 55°C 3.8" lead length, 60Hz, resistive or inductive load	3.0		A
Peak forward surge current I_{FM} (surge) 8.3m seconds single half sine-wave superimposed on rated load (JEDEC method)	150		
Maximum forward voltage V_F at 3.0A, 25°C	1.1	1.7	V
Maximum reverse current at rated reverse voltage T_J = 25°C T_J = 100°C	10.0 500		mA mA
Typical junction capacitance (Note 1) CJ	75	50	pF
Typical junction resistance (Note 2) RθJA	20.0		°C/W
Reverse recovery time I_F = 0.5A, I_R = 1A, I_{rr} = 0.25A	50	75	ns
Operating and storage temperature range	-55 to +150		°C

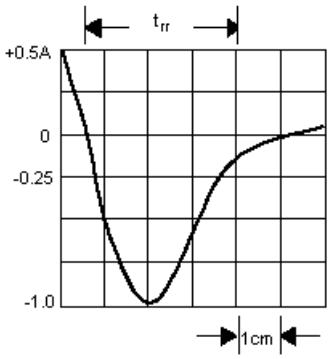
Notes:

1. Measured at 1MHz and applied reverse voltage of 4.0V dc.
2. Thermal resistance from junction to ambient and from junction to lead length 0.375” (9.5mm) PCB mounted.

Rating and Characteristics Curves

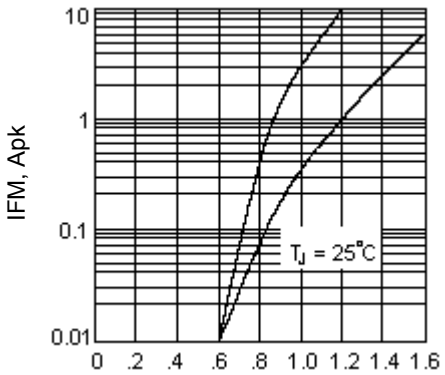


NOTE: 1. Rise Time = 7ns maximum
Input Impedance = 1MΩ, 22pF
2. Rise Time = 10ns maximum
Source Impedance = 50Ω



Set Time
Base For
50ns/cm

Figure 1 - Reverse Recovery Time Characteristics and Test Circuit Diagram



Forward Voltage-VFM(Vpk)

Figure 2 - Forward Characteristics

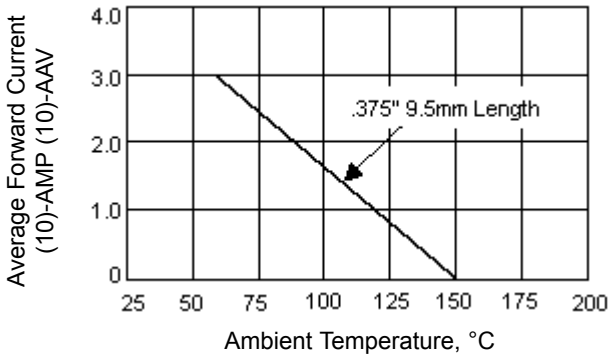


Figure 3 - Forward Current Derating Curve

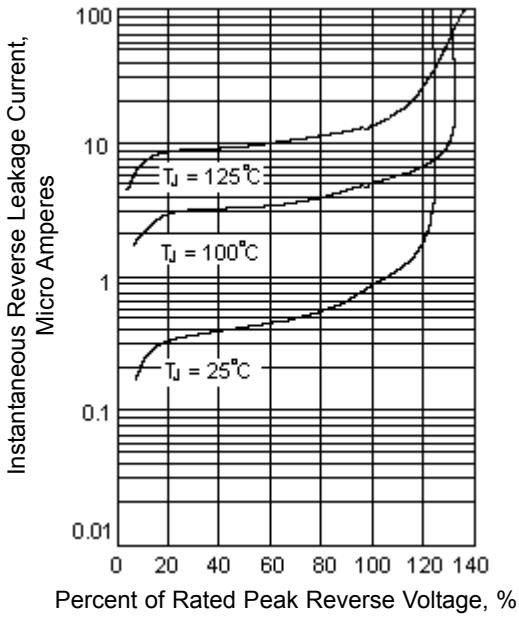


Figure 4 - Typical Reverse Leakage Characteristics

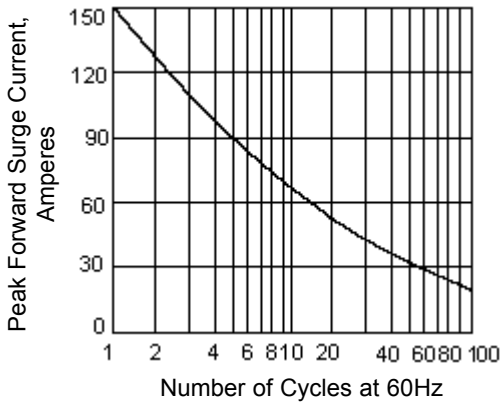


Figure 5 - Peak Forward Surge Current

Specifications

V_{RRM} maximum (V)	$I_{F(av)}$ (A)	I_{FSM} (A)	t_{rr} maximum (ns)	V_F (V) at I_F = 3.0A	Length	Diameter	Part Number
400	3.0	150	50	1.1	9.5	5.3	UF304
800			75	1.7			UF308

Dimensions : Millimetres

Notes:

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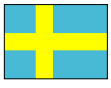
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